

Project Overview:

The U.S. rental housing market is shaped by geographic location, property characteristics, and listing features such as size, amenities, and segmentation. These factors collectively influence pricing patterns across different regions.

This project analyzes approximately 99,000 rental listings across the United States to identify key drivers of rental prices and uncover geographic and market-level trends. Using interactive Power BI dashboards, the analysis explores how property attributes and location impact affordability and rental demand.



Dataset Overview:

Dataset: Apartment for Rent Classified

Source: UCI Machine Learning Repository

Records: 99,826 rental listings

Features: 21 variables including price, bedrooms, bathrooms, square footage, location, amenities, and pet policies.

Why This Dataset?

Large-scale dataset representing real-world U.S. rental housing markets.

Contains both numerical and categorical variables suitable for exploratory analysis and visualization.

Supports investigation of housing affordability, pricing drivers, and geographic trends.

Required data cleaning due to missing values, formatting inconsistencies, and a small number of corrupted records.

Project Objectives

Identify the primary factors influencing rental prices.

Analyze the impact of property characteristics such as bedrooms, bathrooms, and square footage.

Examine geographic differences in rental prices across states and cities.

Evaluate the influence of amenities, pet policies, and listing features.

Explore rental market segmentation across Economy, Standard, Premium, and Luxury listings.

Dataset Cleaning:

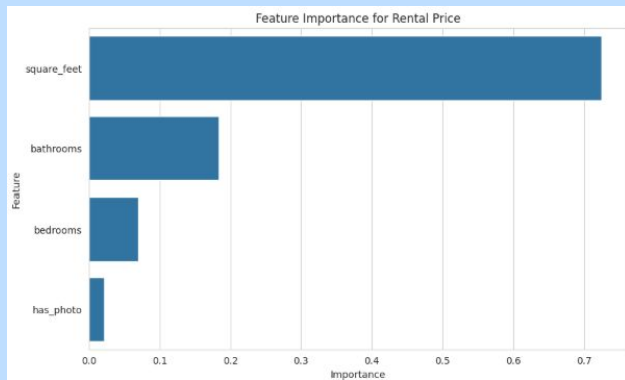
The original dataset contained approximately 100,000 rental listings and required extensive preprocessing due to corrupted records, inconsistent formatting, missing values, and column misalignment issues. This is where I spent most of my time. Using Python, BigQuery, and power query, multiple rounds of data validation and transformation were performed to identify and correct data quality issues. Key cleaning activities included removing invalid and incomplete records, standardizing formats, handling missing values in critical fields such as bedrooms and bathrooms, and eliminating non-essential text fields that caused structural inconsistencies. The final dataset was refined into a reliable and analysis-ready format while preserving valuable features such as amenities for further market analysis.



```
= Table.SelectRows("#Replaced Value", each ([bathrooms] <> null) and ([bedrooms] <> null and [bedrooms] <> 0))
```

```
= Table.TransformColumnTypes("#Promoted Headers",{{"category", type text}, {"amenities", type text}, {"bathrooms", type number}, {"bedrooms", Int64.Type}, {"currency", type text}, {"fee", type text}, {"has_photo", type text}, {"pets_allowed", type text}, {"price", Int64.Type}, {"price_type", type text}, {"square_feet", Int64.Type}, {"cityname", type text}, {"state", type text}, {"latitude", type number}, {"longitude", type number}, {"source", type text}, {"time", type datetime}, {"segment", type text}, {"has_amenities", type text}})
```

Research Questions & Analysis



To identify the factors that most strongly influence rental prices, I developed a Random Forest model. The analysis revealed that square footage was the most important predictor of rental price, highlighting the strong relationship between property size and rental value. Number of bathrooms emerged as the second most influential factor, suggesting that properties with additional bathrooms command higher rental prices due to increased functionality, convenience, and construction costs. These findings indicate that both property size and interior amenities play a significant role in determining rental market value.

Random Forest Regression Model to evaluate "What factors most strongly influence rental prices?"

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split

features = [
    'bedrooms',
    'bathrooms',
    'square_feet',
    'has_photo'
]

data = df_clean[features + ['price']].dropna()

X = data[features]
y = data['price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=.2,
    random_state=42
)

rf = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)

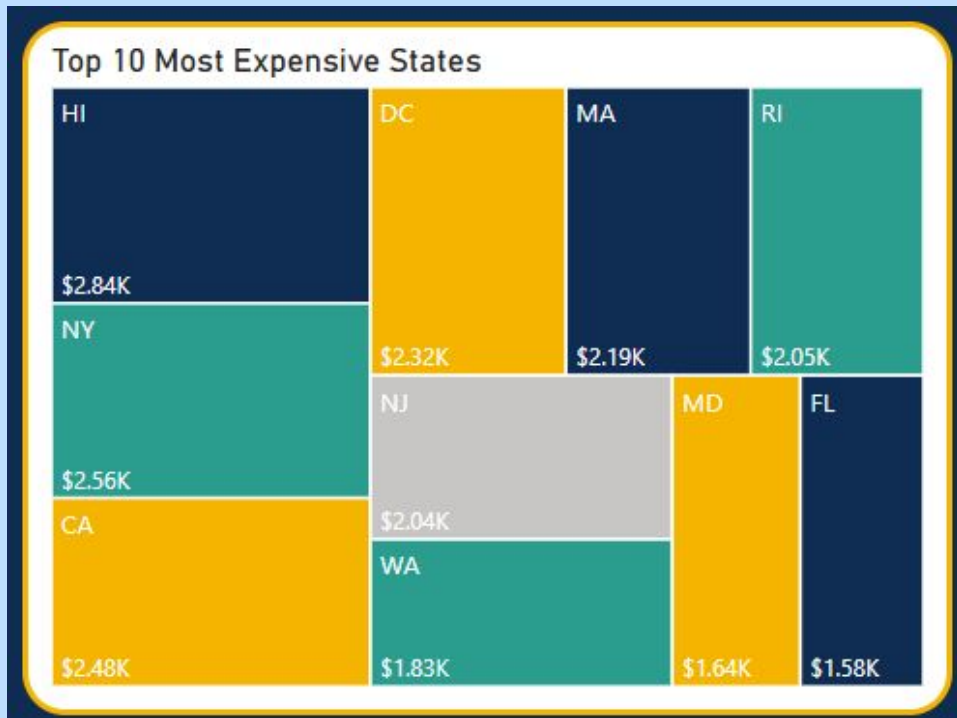
importance = pd.DataFrame({
    'Feature': features,
    'Importance': rf.feature_importances_
}).sort_values(
    'Importance',
    ascending=False
)

sns.barplot(
    data=importance,
    x='Importance',
    y='Feature'
)

plt.title('Feature Importance for Rental Price')
plt.show()
```

Research Questions & Analysis

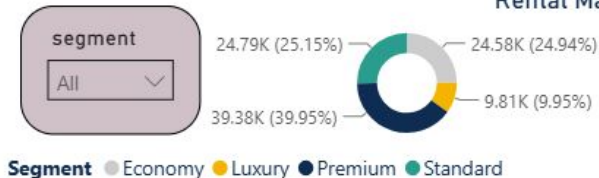
- How do rental prices vary across geographic regions?



This was one of the most effective visuals in the dashboard because it clearly highlights the states with the highest average rental prices. The treemap makes it easy to compare the top 10 most expensive states at a glance, with Hawaii, New York, and California standing out as expected due to their high housing costs. This visualization quickly communicates regional rental price disparities in a way that is both engaging and easy to understand.

Rental Market Overview in the United States

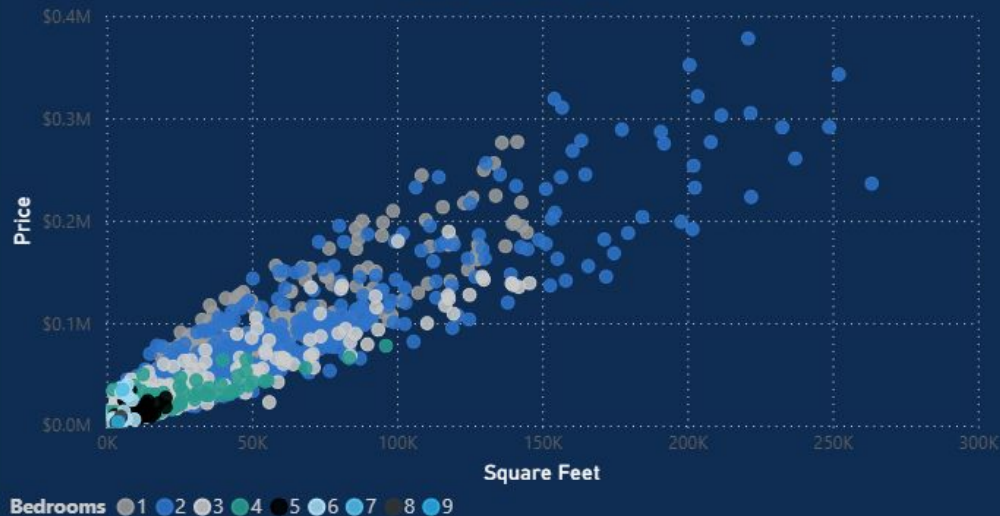
Rental Market Segmentation



Average Monthly Rent by Number of Bathrooms



Square Feet vs Rent



\$1K

Median of Price

99K

Total Listings

958.41

Average Square Feet

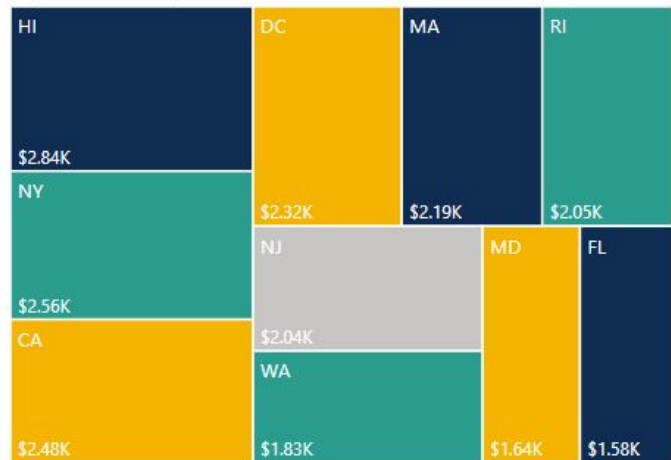
\$1.53K

Average Rent

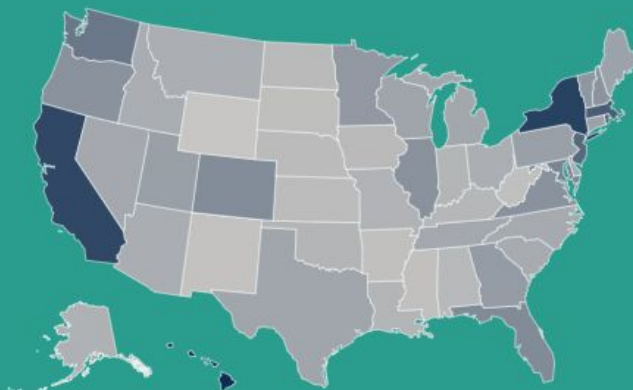
Average Rent by Segment



Top 10 Most Expensive States



Rental Prices Across the United States



Average Apartment Rent by State



Rental Prices by Location



Geographic Market Analysis

Top 10 Most Expensive Cities



DE

Lowest Rent State

CA

Highest Rent State